



Benjamin BILLOT

Researcher at Inria

Inria, Epione team

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Experience

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|--|--------------------------------|---|
| 2025 - now | Researcher | Epione team, Inria Sophia Antipolis, France |
| Permanent researcher of the Epione team at Inria Sophia Antipolis, where my work focuses on developing machine learning strategies to robustly analyse medical images. | | |
| 2022 - 2024 | Postdoctoral researcher | MIT, USA |
| Equivariant and denoising CNNs to decouple spatial and intensity features for motion tracking in fetal MRI
Advisor: Polina Golland | | |
| 2018 | Data Scientist | Founders Factory, London, UK |
| Developed R-CNN for automated navigation of HTML pages | | |
| Summer 2016 | Research assistant | INSERM, France |
| Physics-based generative models in low data regime: segmentation of cortical microscopies.
Advisors: Thérèse Jay and Marco Pompili | | |

Education

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|---|---------------------------------------|-------------------------------|
| 2022 | PhD in medical image computing | University College London, UK |
| Thesis: Bridging generative models and CNNs for domain-agnostic segmentation of brain MRI
Advisor: Juan Eugenio Iglesias | | |
| 2017 | MSc in biomedical engineering | Imperial College London, UK |
| Thesis: Physics-based generative models in low data regime: segmentation of cortical microscopies.
Advisor: Anil Bharath | | |
| 2016 | Diplôme d'ingénieur | CentraleSupélec, France |
| Project: Cancellation of respiratory motions for accurate chest CT reconstruction
Advisor: Elisabeth Lahalle | | |

Honours and awards

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| 2025 | IdEx Attractivity Package |
| 50,000€ awarded by IdEx Université Côte d'Azur | |
| 2022-2025 | Outstanding reviewer |
| MICCAI 2022, MICCAI 2023, MICCAI 2024, MICCAI 2025 | |
| 2019-2024 | Short-listed for best paper award |
| MIDL 2019, MIDL 2020, MIDL 2024 | |
| 2019 | CMIC platform grant |
| £4,000 awarded for a 6-month visit at MIT as a PhD student | |

Summary of publications

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|----|----------------------------|---------------------------------------|
| 19 | Journal articles | 4 as first author, 4 as second author |
| 20 | Conference articles | 7 as first author, 5 as second author |

Software releases

SynthSeg	Domain-agnostic segmentation of brain scans Python/TensorFlow	GitHub FreeSurfer Matlab	github.com/BBillot/SynthSeg surfer.nmr.mgh.harvard.edu/fswiki/SynthSeg mathworks.com/products/matlab.html
SynthSR	Tool to turn any brain scan into a 1mm T1 scan Python/TensorFlow	GitHub FreeSurfer	github.com/BBillot/SynthSR surfer.nmr.mgh.harvard.edu/fswiki/SynthSR
Hypo_seg	Segmentation of the hypothalamic subunits Python/TensorFlow	GitHub FreeSurfer	github.com/BBillot/hypothalamus_seg surfer.nmr.mgh.harvard.edu/fswiki/HypothalamicSubunits
EquiTrack	Motion Tracking in fetal MRI time series Python/PyTorch	GitHub	github.com/BBillot/EquiTrack

Mentoring

2025 - now	Huiyu Li – Postdoctoral researcher Robust AI detection tool for lung cancer prognosis Co-supervised with Hervé Delingette (EPIONE) and Charles Marquette (IHU Nice)	Inria, France
2025 - now	Alix de Langlais - PhD student 3D modelling of proximal humerus fractures for preoperative planning and intraoperative assistance Co-supervised with Hervé Delingette (EPIONE) and Marc-Olivier Gauci (CHU Nice)	Inria, France
2025 - now	Manasi Kattel - PhD student Deep learning methods for the analysis and exploitation of ultrasound images CIFRE thesis with KOELIS, co-supervised with Nicholas Ayache (EPIONE) and Hervé Delingette (EPIONE)	Inria, France
2025 - now	Federica Facente - PhD student Biomedical models for CT/X-ray image analysis - application to image-guided surgical robotics CIFRE thesis with CARANX MEDICAL, co-supervised with Nicholas Ayache (EPIONE) and Hervé Delingette (EPIONE)	Inria, France
2023 - now	Ramya Muthukrishnan - PhD student Equivariant networks for 3D fetal head pose estimation in motion-aware MRI acquisition at high-resolution Co-supervised with Polina Golland (MIT)	MIT, USA
2024	Sebo Diaz - PhD student Landmark detection for fetal motion tracking in whole-uterus MRI time-series Co-supervised with Elfar Adalsteinsson (MIT) and Polina Golland (MIT)	MIT, USA
2019	Bo Hyun Song - MSc student Simulation of histological artefacts in medical images Co-supervised with Juan Eugenio Iglesias (UCL)	University College London, UK

Engagement with the scientific community

EVENT ORGANISATION		
2023-2024	4th and 5th Boston medical Imaging Workshops Head of the organisation committee	
REVIEWING		
Area Chair	MIDL (2024, 2025)	
Reviewer	Journals	IEEE transactions on Medical Imaging, Medical Image Analysis, Nature Communications, Nature Biomedical Engineering, IEEE transactions on Pattern Analysis and Machine Intelligence, Scientific Data, NeuroImage, Imaging Neuroscience, Human Brain Mapping, Frontiers in Neuroscience, Journal of Neurology, Machine Learning for Biomedical Imaging
	Conferences	MICCAI (2021-2025), MIDL (2022, 2023), IPMI (2023), DGM4H NeurIPS Workshop (2023)
	Workshop proposals	EurIPS (2025)
PRESS		
2023	PhysicsWorld press release <i>AI creates high-resolution brain images from low-field strength</i> by Cynthia E. Keen	

TEACHING

2018 **Introductory Mathematics for Computer Science** (10x1h30) University College London, UK

PANELLIST

2025 **Best practices for MICCAI reviews and rebuttals** MICCAI student board webinar, virtual
2024 **How to write an award-winning paper?** MIDL Winter Workshop, virtual

SOCIETY MEMBERSHIP

2020 – now **MICCAI member**

Invited presentations

Network conditioning for synergistic learning on partial supervision

202 Sabuncu Lab seminars Cornell Tech, New York City, USA
2024 LEMoN group lab seminars Harvard Medical School, Boston, USA

SE(3)-equivariant and noise-invariant 3D rigid motion tracking in brain MRI

2024 ARAMIS seminars Inria, Paris, France
 IMAGES seminars Télécom Paris, France
2023 EPIONE seminars Inria, Nice, France
 MICS seminars CentraleSupélec, Paris, France

Robust segmentation of heterogeneous clinical brain MRI scans

2025 Statlearn'25 spring school Inria, Nice, France
 3IA Côte d'Azur seminars 3IA, Nice, France
2024 Martinos center lab seminars Harvard Medical School, Boston, USA
 CMIC-WEISS seminars UCL, London, UK
2023 LEMoN group lab seminars Harvard Medical School, Boston, USA

SynthSeg: domain-randomisation for segmentation of brain MRI of any contrast and resolution

2022 Biomedical imaging and analysis seminars MIT, Boston, USA
2021 CMIC-WEISS seminars UCL, London, UK
2020 LCN group seminars Harvard Medical School, Boston, USA
 Bio-imaging symposium Imperial College London, UK